Naive Bayes Classifier | Part 4 | Bayes Theorem in Probability

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1 What is Bayes' Theorem?

Bayes' Theorem is a mathematical rule that lets us **update our belief about an event** when we get new evidence.
Formula:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Where:

- P(A|B) → **Posterior** = Probability of event **A** given evidence **B**.
- P(B|A) → **Likelihood** = Probability of observing evidence **B** if **A** is true.
- P(A) → **Prior** = Our initial belief about **A** (before evidence).
- P(B) → **Evidence** = Total probability of observing **B**.

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2 Real-Life Intuition

Think of a doctor diagnosing a disease:

- **Prior (P(A))**: What is the general chance of having the disease (before test)?
- **Likelihood (P(B|A))**: If the patient has the disease, what is the chance the test comes positive?
- **Evidence (P(B))**: Overall chance of a positive test result (whether disease is there or not).
- **Posterior (P(A|B))**: Given the test result is positive, what is the updated probability that the patient has the disease?

👉 Bayes helps doctors make **better predictions with evidence**.

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3 Example

Let's say:

- 1% of people have a rare disease → $P(Disease) = 0.01.$
- If diseased, test shows positive 99% of the time → $P(Positive|Disease) = 0.99.$
- If healthy, test shows positive 5% of the time (false alarm) → $P(Positive|NoDisease) = 0.05.$

Now, if a patient's test is **positive**, what's the chance they really have the disease?

$$P(Disease|Positive) = \frac{P(Positive|Disease) \cdot P(Disease)}{P(Positive)}$$

Where:

$$P(Positive) = (0.99)(0.01) + (0.05)(0.99) = 0.0099 + 0.0495 = 0.0594$$

So:

$$P(Disease|Positive) = \frac{0.99 \cdot 0.01}{0.0594} = \frac{0.0099}{0.0594} \approx 0.166$$

👉 Even after a **positive test**, chance of disease is **only ~16.6%**. Surprising but true!

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4 Why is this important for Naive Bayes?

Naive Bayes classifier is built directly from this theorem:

$$P(Class|Features) = \frac{P(Features|Class) \cdot P(Class)}{P(Features)}$$

- We calculate probability of each **class** given the observed **features**.
- Choose the class with the **highest posterior probability**.

✅ Quick Recap

- Bayes' Theorem updates probabilities using evidence.
- Formula: $P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$
- Key terms: **Prior, Likelihood, Evidence, Posterior**.
- Example shows how probabilities can be very different after evidence.
- **Naive Bayes classifier** applies Bayes' Theorem with the assumption that features are **independent given the class**.
